Thematic Analysis Tool for Health Providers Specification

Problem Statement:

Patient and public survey data are crucial for shaping health services, but analysing qualitative (free-text) feedback requires significant manual effort. While existing AI tools can generate themes rapidly, they often act as "black boxes," failing to provide the transparency needed to verify how conclusions were drawn. This lack of transparency leads to the following issues:

- A high risk of unchecked bias in data interpretation.
- Reduced stakeholder and public trust in AI-generated insights.
- Difficulty in tracing high-level themes back to the original patient voices.
- Limited usefulness of the data for safe, evidence-based service design.

Proposed Solution:

By initially injecting quotes into generated clustered *themes*, an AI model will learn your data's patterns & initial themes to suggest clustering of future survey responses, with a human-in-the-loop (HITL) capable of accepting, rejecting, or modifying AI suggestions. This tool will enable transparency and evidence-based insights to safely shape future health services.

The system will:

- **Process** large volumes of unstructured qualitative survey feedback.
- **Identify** recurring themes, concerns, and ideas from the public/patient voice.
- **Cross-reference** generated themes with underlying metadata (e.g., demographics, location).
- **Provide** a transparent audit trail linking every theme back to specific, anonymised patient quotes.

Key Features

- **Transparent Thematic Analysis:** Automatically identifies and categorises recurring themes while explicitly showing the rules or data weightings used to form them. Also offers a HITL playground for a visualisation tool and to verify the automatic clustering.
- **Quote Traceability (Evidence Linking):** Allows users to click on any AI-generated theme to instantly read the exact, original survey responses that generated it, eliminating the "black box" effect.
- **Metadata Integration:** Overlays qualitative themes with quantitative metadata (e.g., age, region, health condition) to highlight specific demographic trends and disparities.

- **Bias Detection & Mitigation:** Flags potential biases in the data (e.g., alerting the user if a major theme is driven by a single demographic group).
- **Secure, In-House Processing:** Keeps sensitive patient feedback entirely within the organisation's secure IT environment, ensuring strict compliance with UK GDPR and NHS data protection standards.
- **Exportable Insight Reports:** Generates clear, structured reports and visualisations for health service planners and policymakers.

Summary:

In conclusion, this transparent AI analysis tool transforms overwhelming volumes of qualitative patient feedback into actionable, evidence-based insights. By solving the "black box" problem and linking themes directly to demographic metadata and original quotes, the tool builds trust, reduces bias, and empowers decision-makers to design health services that accurately reflect the true public voice.

